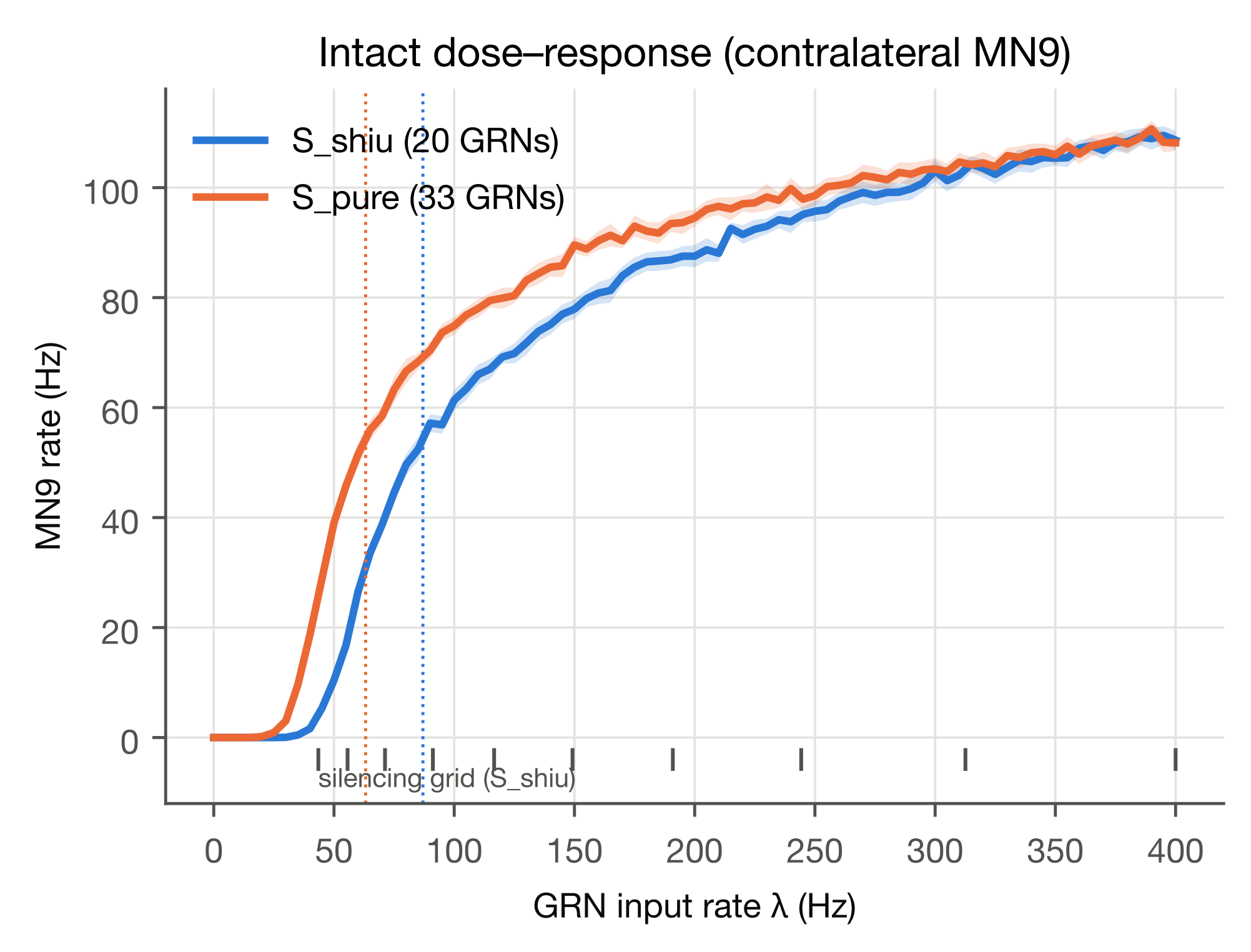
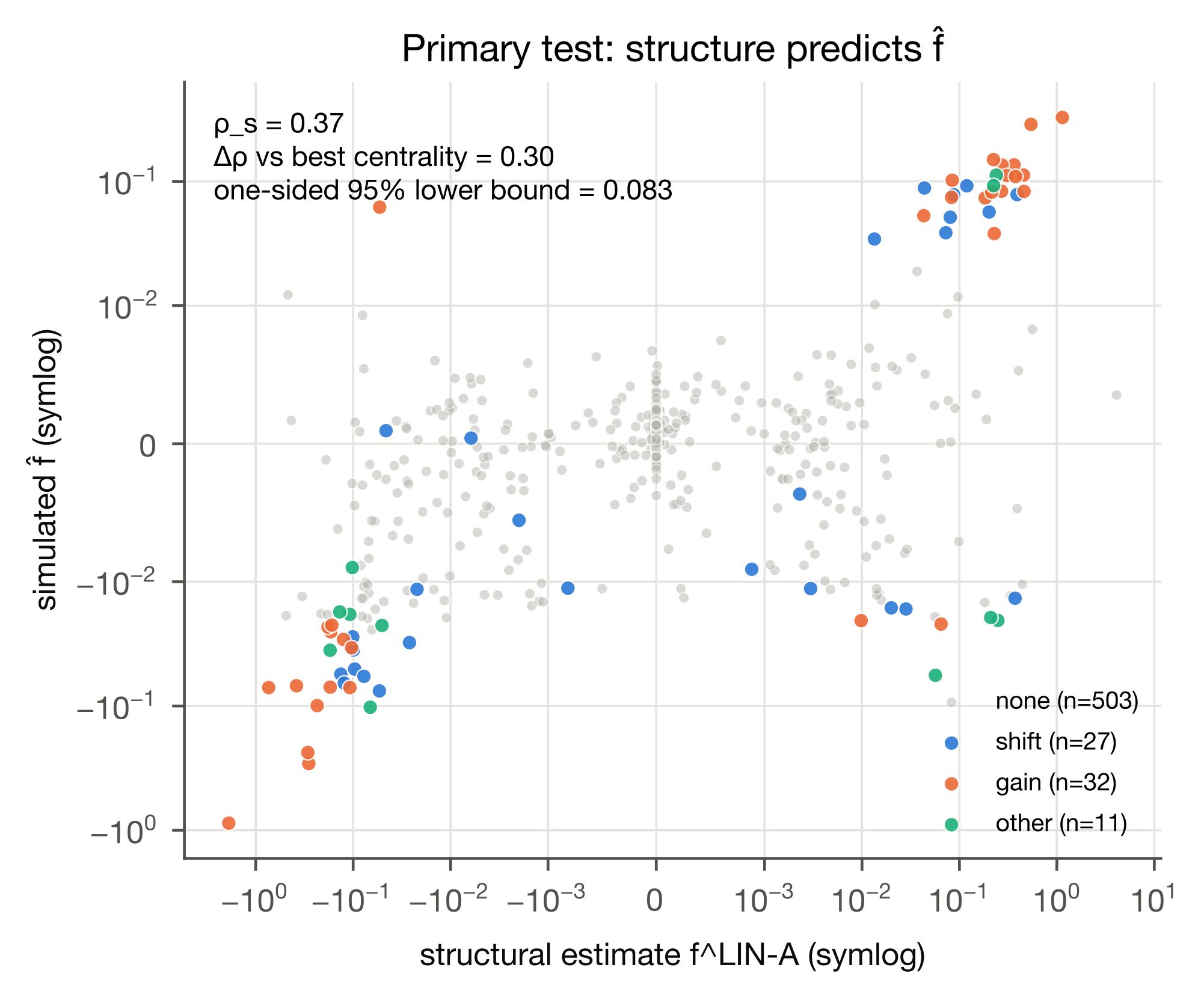
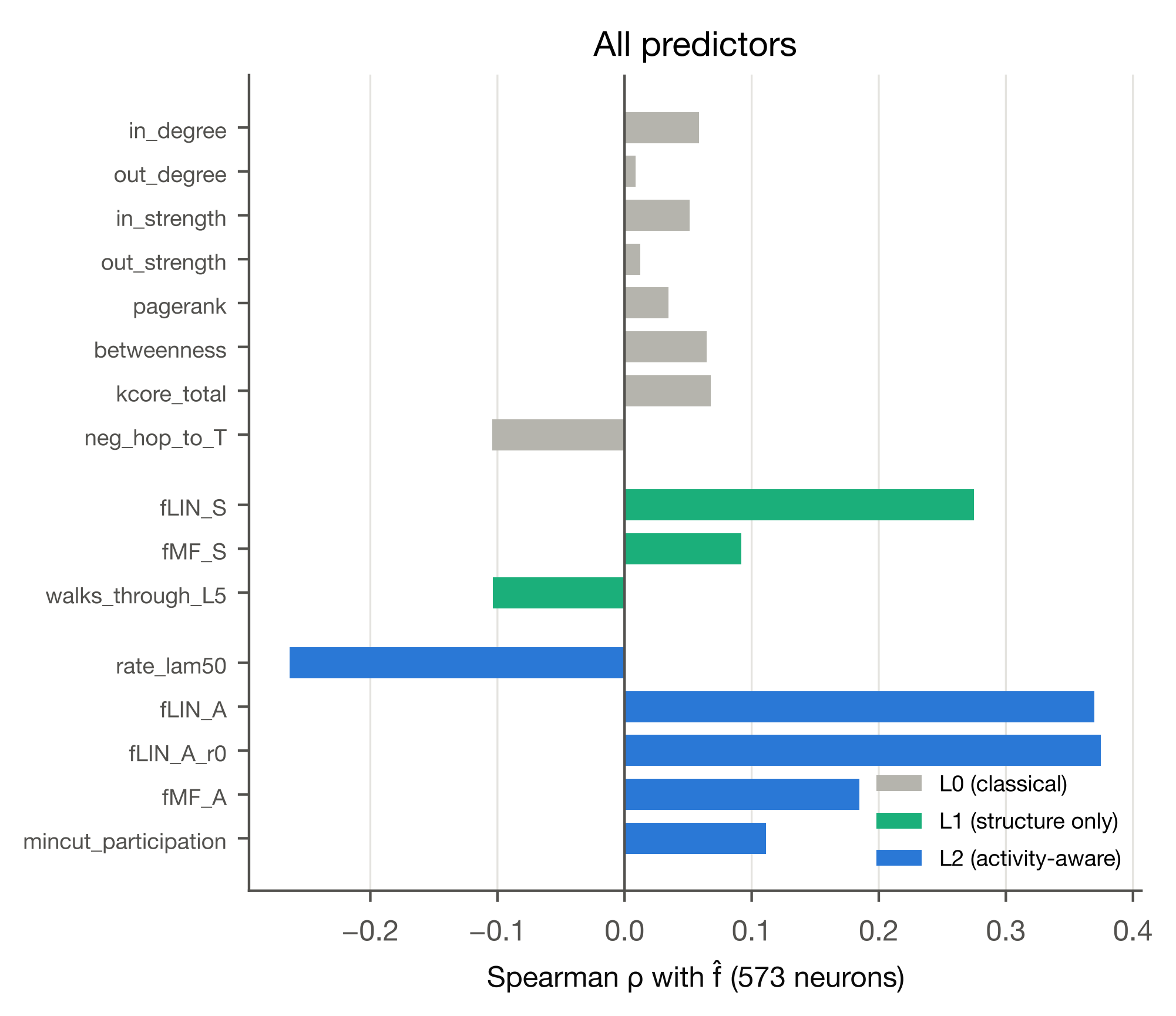
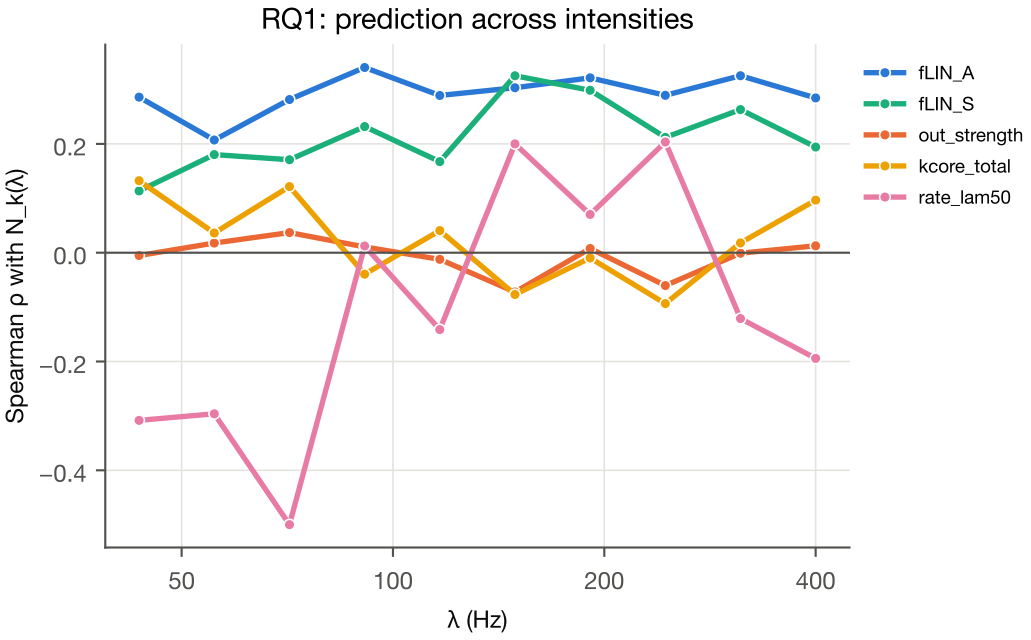
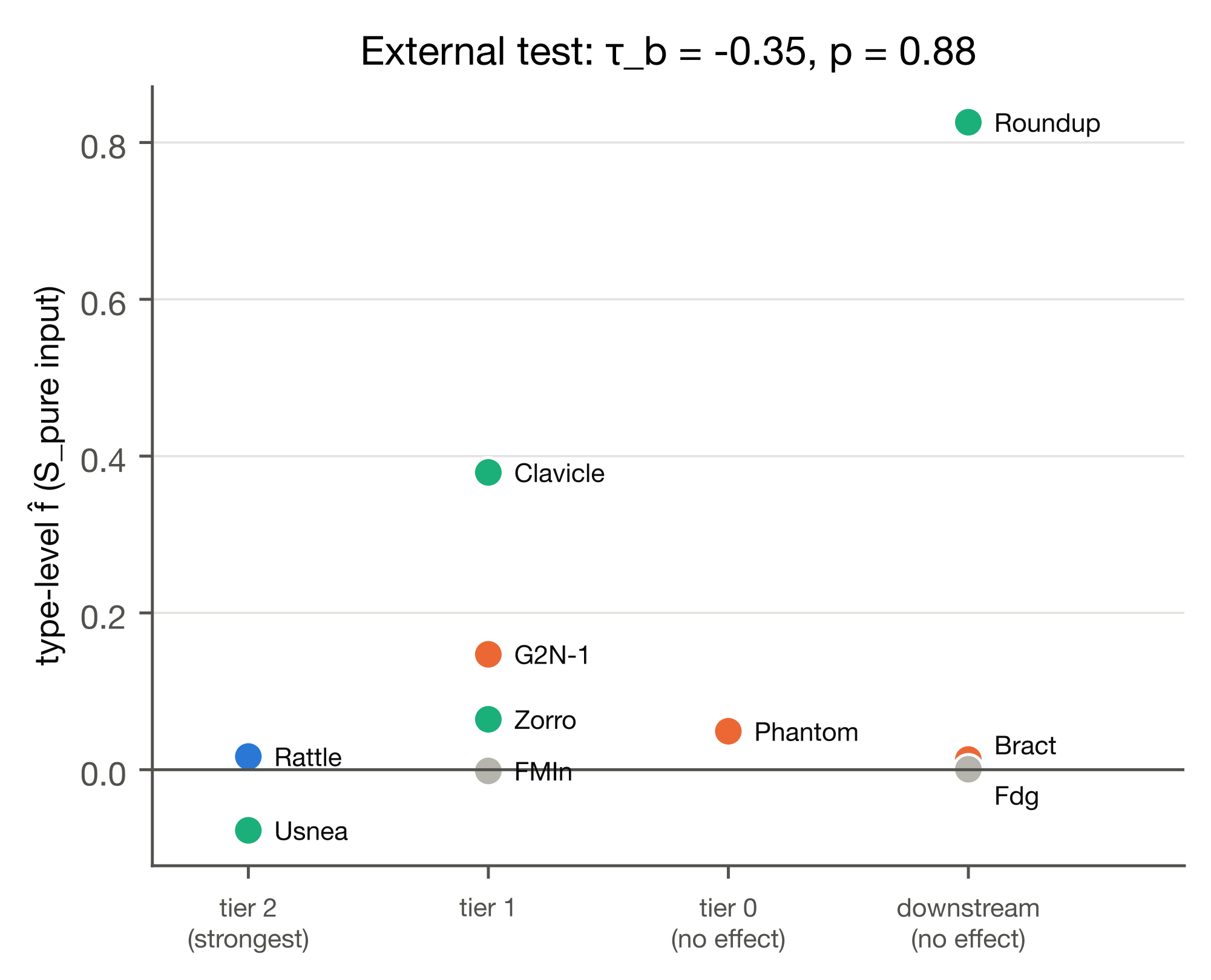
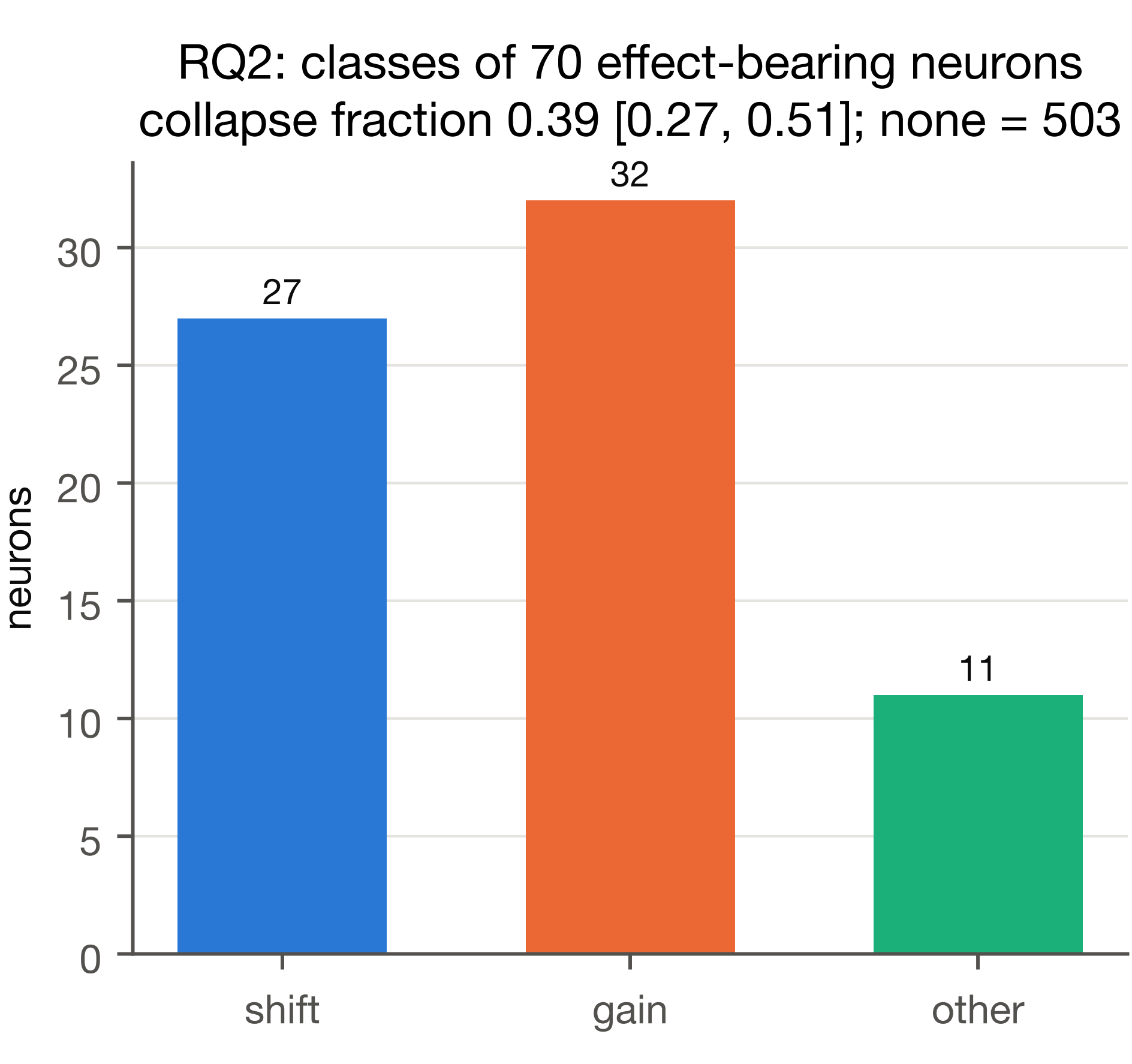

Intact dose–response (contralateral MN9)
S_shiu (20 GRNs)
S_pure (33 GRNs)
MN9 rate (Hz)
0
20
40
60
80
100
silencing grid (S_shiu)
0
50
100
150
200
250
300
350
400
GRN input rate λ (Hz)

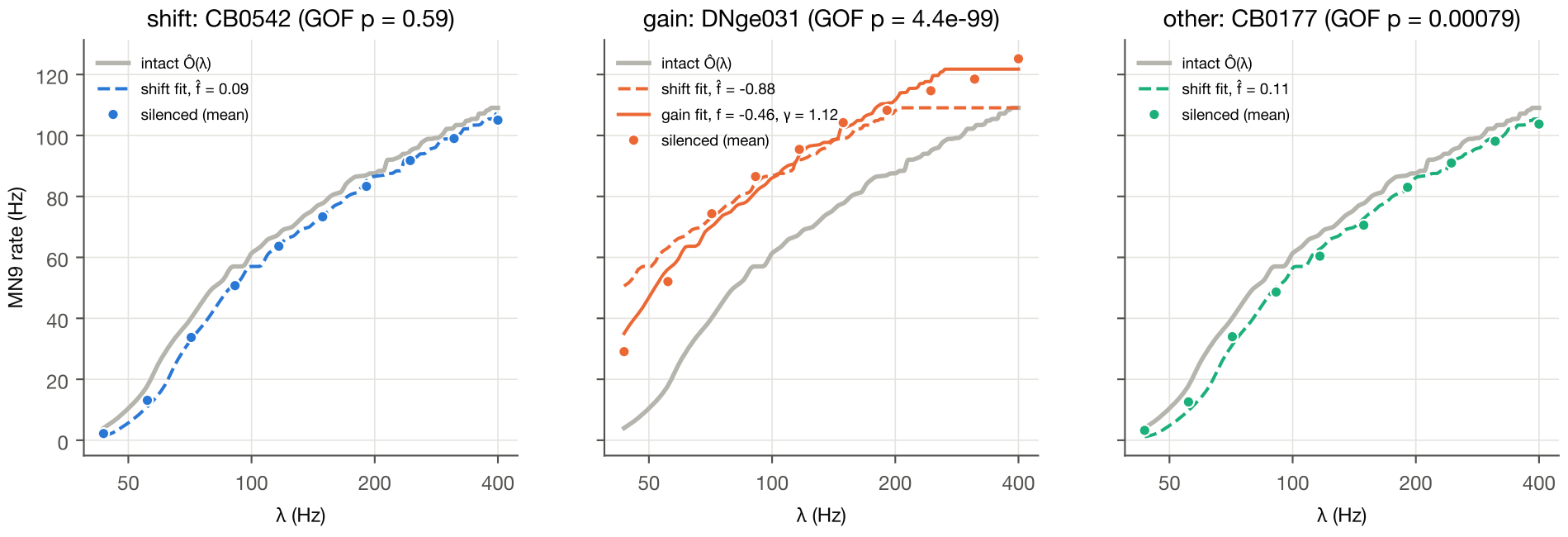

Primary test: structure predicts f̂
ρ_s = 0.37
Δρ vs best centrality = 0.30
one-sided 95% lower bound = 0.083
simulated f̂ (symlog)
structural estimate f^LIN-A (symlog)
none (n=503)
shift (n=27)
gain (n=32)
other (n=11)


All predictors
in_degree
out_degree
in_strength
out_strength
pagerank
betweenness
kcore_total
neg_hop_to_T
fLIN_S
fMF_S
walks_through_L5
rate_lam50
fLIN_A
fLIN_A_r0
fMF_A
mincut_participation
L0 (classical)
L1 (structure only)
L2 (activity-aware)
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Spearman ρ with f̂ (573 neurons)


RQ1: prediction across intensities
Spearman ρ with N_k(λ)
0.2
0.0
−0.2
−0.4
50
100
200
400
λ (Hz)
fLIN_A
fLIN_S
out_strength
kcore_total
rate_lam50


External test: τ_b = -0.35, p = 0.88
type-level f̂ (S_pure input)
0.8
0.6
0.4
0.2
0.0
Roundup
Clavicle
G2N-1
Zorro
Phantom
Bract
Rattle
FMln
Fdg
Usnea
tier 2
(strongest)
tier 1
tier 0
(no effect)
downstream
(no effect)


RQ2: classes of 70 effect-bearing neurons
collapse fraction 0.39 [0.27, 0.51]; none = 503
neurons
0
5
10
15
20
25
30
27
32
11
shift
gain
other


Example silencing curves (largest |f̂| per class)
shift: CB0542 (GOF p = 0.59)
gain: DNge031 (GOF p = 4.4e-99)
other: CB0177 (GOF p = 0.00079)
intact Ô(λ)
shift fit, f̂ = 0.09
silenced (mean)
shift fit, f̂ = -0.88
gain fit, f = -0.46, γ = 1.12
silenced (mean)
intact Ô(λ)
shift fit, f̂ = 0.11
silenced (mean)
MN9 rate (Hz)
0
20
40
60
80
100
120
50
100
200
400
λ (Hz)